

## Bill of Materials - GridStore-D3

VBF-T2R1-BOM-01 | t2\_r1 grid energy storage | 2026-08-25 | values mechanically sourced from parameter set / simulation outputs

### Bill of materials (dual caliber: g/cell and kg/kWh)

| Component                                                                             | Volume fraction / basis                         | Mass g/cell | kg/kWh | Source / formula                                                |
|---------------------------------------------------------------------------------------|-------------------------------------------------|-------------|--------|-----------------------------------------------------------------|
| Positive electrode active material (NMC811)                                           | 0.665 (param set)                               | 0.0168      | 0.0009 | th 75.6um x area 0.1027m2 x vf 0.665 x rho 3262 (Chen2020 dump) |
| Positive electrode conductive additive (CB)                                           | 4 wt% of active (lit. default, NMC811 92:4:4)   | 0.0007      | 0.0000 | literature default, annotated                                   |
| Positive electrode binder (PVDF)                                                      | 4 wt% of active (lit. default)                  | 0.0007      | 0.0000 | literature default, annotated                                   |
| Negative electrode active material (graphite)                                         | 0.75 (param set)                                | 0.0109      | 0.0006 | th 85.2um x area x vf 0.75 x rho 1657 (Chen2020 dump)           |
| Negative electrode conductive additive (CB)                                           | 2 wt% of active (lit. default, graphite 96:2:2) | 0.0002      | 0.0000 | literature default, annotated                                   |
| Negative electrode binder (CMC/SBR)                                                   | 2 wt% of active (lit. default)                  | 0.0002      | 0.0000 | literature default, annotated                                   |
| Separator                                                                             | porosity 0.47                                   | 0.0003      | 0.0000 | th 12um x area x (1-0.47) x rho 397 (Chen2020 dump)             |
| Electrolyte                                                                           | pore volume x 1.2 g/cm3 (lit. density)          | 8.6219      | 0.4672 | pore volume 7.185 cm3 x 1.2 g/cm3; fill factor 100%             |
| Positive current collector (Al)                                                       | -                                               | 0.0044      | 0.0002 | th 16um x area x rho 2700 (Chen2020 dump)                       |
| Negative current collector (Cu)                                                       | -                                               | 0.0110      | 0.0006 | th 12um x area x rho 8960 (Chen2020 dump)                       |
| Enclosure                                                                             | Not modeled                                     | Not modeled |        | beyond simulation boundary                                      |
| Tabs                                                                                  | Not modeled                                     | Not modeled |        | beyond simulation boundary                                      |
| TOTAL (BOM caliber: incl. lit-default additive/binder + electrolyte)                  |                                                 | 8.667       | 0.4697 | sum of rows; total energy 18.4542 Wh                            |
| TOTAL cell mass (contract caliber: layers only, electrolyte/binder/additive excluded) |                                                 | 39.633      | 2.1477 | cell/r6_d3_energy.json:mass_kg                                  |